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LOGOUT

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Tipo laurea: Magistrale

Titolo provvisorio: Leveraging Non-Unitary Protocols to Enhance First Detection

Tipo di attività: Teorica

Abstract: The idea that physical systems are fundamentally information-bearing was proposed by Landauer in 1991 and it stuck in the scientific community ever since. With the recent development of quantum computation and algorithms, a versatile framework emerged: quantum walks can be readily thought of as a generalization of classical random walks, but a perspective shift also allows us to study any time-independent Hamiltonian dynamics as a quantum walk on a graph, as long as the Hermiticity and the locality of the Hamiltonian match the graph's topology. Quantum walks are no exception to Feynman's interpretation of quantum mechanics: multiple paths leading to the same outcome interfere, and a non-trivial consequence of this is that quantum walks can outperform their classical counterparts in many applications. There is never a shortage of questions coming from the measurement phase in quantum mechanics, which breaks the deterministic nature of the Schrödinger evolution and introduces the randomness ingredient. In this work, we address the problem of state-transfer over a network, which recently piqued the interest of researchers, both in the unitary regime and from the point of view of a projective measurement as well. In particular, it was recently shown that topology and chirality of graphs can be harnessed to make transport more efficient, and also that a stroboscopic measurement protocol can strongly influence the statistics for the detection of a particle (or many) on a target space. Furthermore, we outline the recent experimental attempts for demonstrations of such protocols, exposing the possible challenges faced by experimentalists on present day devices.

Keywords: Quantum Dynamics, Quantum Measurements, Quantum Transport,

Quantum Walks, Conditional Measurements

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